

Business Calculus (MATH 2350)  
Summer 2018  
Thursday, 06/14/2018  
Quiz 6: Section 12.1, 12.2

Name (Print): \_\_\_\_\_

Score: \_\_\_\_\_ / 110

## Section 12.1

1. 10 points If  $f(x) = \frac{2}{x} - \frac{2}{x^2}$ , find the intervals on which  $f(x)$  is increasing.

First, we find  $f'(x)$ :

$$f'(x) = -2x^{-2} + 4x^{-3}.$$

Then, we find the roots of  $f'(x)$ :

$$\begin{aligned} f'(x) &= 0 \\ -2x^{-2} + 4x^{-3} &= 0 \\ 4x^{-3} &= 2x^{-2} \\ 4x^{-1} &= 2 \\ x &= 2. \end{aligned}$$

Then, we also note that  $x = 0$  is a vertical asymptote of  $f'(x)$ . From this, we then construct our sign chart, finding that

$$f'(x) < 0 \text{ on } (-\infty, 0) \cup (2, \infty), \quad f'(x) > 0 \text{ on } (0, 2).$$

Thus, as  $f(x)$  is increasing when  $f'(x) > 0$ , we conclude that  $f(x)$  is increasing on  $(0, 2)$ .

2. A manufacturer incurs the following costs in producing  $x$  boomerangs in one day, for  $0 < x < 120$ :

Fixed costs: \$300

unit production cost: \$5 per boomerang

equipment and repairs:  $\$0.002x^3$

So, we yield the cost function  $C(x) = 300 + 5x + 0.002x^3$ .

- (a) 10 points What is the average cost  $\bar{C}(x)$  per boomerang if  $x$  boomerangs are produced in one day?

First, we recall that  $\bar{C}(x) = \frac{C(x)}{x}$ . As such, we may conclude that

$$\bar{C}(x) = \frac{300 + 5x + 0.002x^3}{x}.$$

- (b) 10 points Find the intervals on which  $\bar{C}(x)$  is increasing, and the intervals on which  $\bar{C}(x)$  is decreasing.

First, we calculate  $\bar{C}'(x)$ :  $\bar{C}'(x) = -300x^{-2} + 0.004x$ .

Then, we find the roots of  $\bar{C}'(x)$  by calculating:

$$\begin{aligned}\bar{C}'(x) &= 0 \\ -300x^{-2} + 0.004x &= 0 \\ 0.004x &= 300x^{-2} \\ 0.004x^3 &= 300 \Rightarrow x = \sqrt[3]{\frac{300}{0.004}} = \sqrt[3]{75,000}.\end{aligned}$$

Then, we note that  $\bar{C}'(x)$  has a vertical asymptote at  $x = 0$ . Following this, we construct a sign chart for  $\bar{C}'(x)$ , yielding:

$$\bar{C}'(x) < 0 \text{ on } (-\infty, 0) \cup (0, \sqrt[3]{75,000}), \quad \bar{C}'(x) > 0 \text{ on } (\sqrt[3]{75,000}, \infty).$$

Thus, as  $\bar{C}(x)$  is increasing when  $\bar{C}'(x) > 0$  and decreasing when  $\bar{C}'(x) < 0$ , we may conclude that  $\bar{C}(x)$  is increasing on  $(\sqrt[3]{75,000}, \infty)$  and decreasing on  $(-\infty, 0) \cup (0, \sqrt[3]{75,000})$ .

3. 10 points Let  $f(x) = (x-9)e^{-5x}$ . Find the intervals on which  $f(x)$  is increasing, the intervals on which  $f(x)$  is decreasing, and the local extrema.

First, we calculate  $f'(x)$ :

$$f'(x) = (x-9)'e^{-5x} + (x-9)(e^{-5x})' = e^{-5x} - 5(x-9)e^{-5x} = (46-5x)e^{-5x}.$$

Then, we calculate the roots of  $f'(x)$ :

$$\begin{aligned}(46-5x)e^{-5x} &= 0 \\ 46-5x &= 0 \quad (\text{since } e^{-5x} \neq 0 \text{ for all } x) \\ x &= \frac{46}{5}.\end{aligned}$$

Then, we construct a sign chart for  $f'(x)$ , yielding:

$$f'(x) > 0 \text{ on } (-\infty, \frac{46}{5}), \quad \text{and } f'(x) < 0 \text{ on } (\frac{46}{5}, \infty).$$

Thus, as  $f(x)$  is increasing when  $f'(x) > 0$  and decreasing when  $f'(x) < 0$ , we may conclude that  $f(x)$  is increasing on  $(-\infty, \frac{46}{5})$  and decreasing on  $(\frac{46}{5}, \infty)$ .

Moreover, as  $f(x)$  changes from increasing to decreasing at  $\frac{46}{5}$ , we may conclude that  $f(x)$  has a maximum at  $x = \frac{46}{5}$  of  $f(\frac{46}{5}) = \frac{1}{5e^{46}}$ .

4. 10 points If  $f(x) = x^3 + 5x - 24$ , find the intervals on which  $f(x)$  is increasing.

First, we calculate  $f'(x)$ :

$$f'(x) = 3x^2 + 5.$$

Then, we note that  $f'(x)$  has no roots. As such, we test a point – say,  $x = 0$  – and find that  $f'(x) > 0$  on  $(-\infty, \infty)$ . Thus, as  $f(x)$  is increasing when  $f'(x) > 0$ , we may conclude that  $f(x)$  is increasing on  $(-\infty, \infty)$ .

Section 12.2

1. 10 points If  $y = \frac{6}{x} + e^{3x}$ , find  $\frac{d^2y}{dx^2}$ .

First, we calculate  $\frac{dy}{dx}$ :

$$\frac{dy}{dx} = \frac{-6}{x^2} + 3e^{3x}.$$

Then, we calculate  $\frac{d^2y}{dx^2}$ :

$$\frac{d^2y}{dx^2} = \frac{12}{x^3} + 9e^{3x}.$$

2. 10 points If  $f(x) = 4x^6 + 13x + 13$ , find any values of  $x$  for which  $f''(x) = 0$ .

First, we calculate  $f'(x)$ :

$$f'(x) = 24x^5 + 13.$$

Then, we calculate  $f''(x)$ :

$$f''(x) = 120x^4.$$

Finally, we solve  $f''(x) = 0$ :

$$f''(x) = 0$$

$$120x^4 = 0$$

$$x^4 = 0$$

$$x = 0.$$

3. 20 points Sketch a graph of function  $f(x)$  which is continuous on  $(-\infty, \infty)$  and satisfies:

$$f(-2) = 3, \quad f'(0) = 0, \quad f'(-2) = 0, \quad f''(-1) = 0,$$

$$f'(x) < 0 \text{ on } (-2, 0), \text{ and } f'(x) > 0 \text{ on } (0, \infty) \text{ and } (-\infty, -2).$$

(See examples given in class)

4. 10 points If  $f(x) = 4x^6 + 13x + 13$ , find any values of  $x$  for which  $f''(x) = 0$ .

First, we calculate  $f'(x)$ :

$$f'(x) = 24x^5 + 13.$$

Then, we calculate  $f''(x)$ :

$$f''(x) = 120x^4.$$

Finally, we solve  $f''(x) = 0$ :

$$f''(x) = 0$$

$$120x^4 = 0$$

$$x^4 = 0$$

$$x = 0.$$

5. 10 points If  $f(x) = x^3 + 18x^2$ , find the  $x$  and  $y$  coordinates of all inflection points.

First, we calculate  $f'(x)$ :

$$f'(x) = 3x^2 + 36x.$$

Then, we calculate  $f''(x)$ :

$$f''(x) = 6x + 36.$$

Then, we find the roots of  $f''(x)$ :

$$f''(x) = 0$$

$$6x + 36 = 0$$

$$x = \frac{-36}{6} = -6.$$

Then, we construct a sign chart for  $f''(x)$ , yielding:

$$f''(x) < 0 \text{ on } (-\infty, -6), \quad f''(x) > 0 \text{ on } (-6, \infty).$$

As an inflection point occurs wherever  $f''(x)$  changes sign, we then conclude that  $(-6, f(-6))$  is an inflection point of  $f(x)$ .